

# REWEAR-FUSED: Wet-Lab Validation and Experimental De-Risking Roadmap (In-Silico → Pilot)

## TL;DR

- **The single most credible de-risking path is a STTR-funded academic co-validation:** express the designed urethanase (Engine B) in *E. coli*, prove activity on a p-nitrophenyl/UMG-SP2-style carbamate surrogate, then show the matched designed fiber (Engine A) releases its diamine/diol monomers on enzyme incubation (HPLC/LC-MS) — the "matched-pair" experiment is the whole platform's keystone and the one result that de-risks all three engines at once.
- **The cheapest, most convincing FIRST experiment** is a ~1-week, ~\$1-3K colorimetric/fluorometric carbamate-hydrolysis assay on a small batch of expressed designs (cell-free or small-scale *E. coli*), because Lauko et al. (*Science*, Feb 13 2025) showed de-novo serine hydrolases can be found "in screens of as few as 20 designs" at ~10% activity hit rates. nih
- **Realistic budget/timeline:** ~\$150-350K and 12 months to first matched-pair wet-lab proof, best executed via an NSF or DOE STTR with a university partner plus NREL BOTTLE access. The Cox Cleantech Residency (\$10K stipend, 6 weeks, equity-free) is a *bridge* — use it to file a provisional patent and submit the NSF Project Pitch, not as a research budget.

## Key Findings

- **De-novo enzyme design has credible, published hit rates.** Lauko et al. (*Science* 388, eadu2454, published Feb 13, 2025; Baker Lab / Institute for Protein Design, Univ. of Washington) reported that in-silico preorganization filtering "enabl[ed] identification of new catalysts in screens of as few as 20 designs," and the best design, momi120-103, achieved "a kcat for 4MU-PhAc of  $0.057 \text{ s}^{-1}$ , Km of  $0.26 \mu\text{M}$ , and a kcat/Km of  $2.2 \times 10^5 \text{ M}^{-1} \text{ s}^{-1}$ ." This is the empirical foundation for claiming Engine B is screenable cheaply. nih
- **The UMG-SP2 urethanase the platform anchors on is fully characterized in the public literature** — crystal structures (PDB 8WDW), the Ser190-Ser166cis-Lys91 catalytic triad, validated model substrates, and a QM/MM-resolved two-stage (acylation/deacylation) mechanism — giving Engine B a benchmark and ready-made assays.
- **The validation toolchain is commercially turnkey:** Twist clonal genes (NGS-verified; Express in as few as 4-7 business days, standard 10-15), *E. coli* BL21(DE3)/pET expression, IMAC/SEC purification, colorimetric carbamate assays, and HPLC/LC-MS for monomer release.
- **Competitors validate the thesis and define the FTO landscape.** Carbios "owns 116 titles worldwide representing 30 patent families" on enzymatic PET recycling; Samsara Eco and Protein Evolution are extending AI-designed enzymes to PU/nylon. None yet claims a co-

*designed matched enzyme+fiber pair* — REWEAR-FUSED's defensible white space.

Waste360

- **Non-dilutive money is abundant and aligned:** NSF SBIR/STTR (~\$305K Phase I), DOE BOTTLE/bioenergy, and textile-circularity programs (Fashion for Good, Global Fashion Agenda Trailblazer).

## Details

### PART 1 — Engine B: Enzyme Wet-Lab Validation Pathway

#### (a) Sequence → measured activity workflow.

1. **Gene design & codon optimization.** Free tools: GenScript GenSmart, Twist codon optimization (complimentary). Optimize for *E. coli* BL21 codon usage, remove internal restriction sites, append an N-/C-terminal His<sub>6</sub> tag.
2. **Synthetic gene ordering.** Twist Bioscience clonal genes are NGS-verified; standard turnaround averages 10–15 business days and the Express Genes service ships NGS-verified clonal genes in as few as 4–7 business days (Twist ran a period of 4-day academic Express at no premium). GenScript gene synthesis has a \$99/gene floor (standard) and \$129/gene (express minimum). IDT and Twist are the other major vendors. For a ~1 kb urethanase ORF expect roughly \$150–400 per construct delivered in a vector (Twist per-bp pricing ~\$0.07–0.10/bp is an industry estimate, not re-verified here). [Twist Bioscience + 2](#)
3. **Vector/host.** pET-28a(+) (T7 promoter, His-tag) in *E. coli* BL21(DE3) is the default; SHuffle/Origami for disulfides; Rosetta for rare codons. Sino Biological notes that swapping a His-tag for an MBP fusion can rescue solubility (inclusion-body → soluble) — relevant because amidase-family folds can be expression-challenging.
4. **Expression & purification.** IPTG induction or autoinduction; sonication lysis; IMAC (Ni-NTA) capture; SEC polish. Academic core pricing (e.g., UNC Protein Expression & Purification core) runs a few hundred dollars per 4 L *E. coli* culture including transformation and induction (exact rate not re-verified).
5. **Cost/time per design (in-house/academic):** roughly \$300–800 and 3–5 weeks from gene to purified protein.

#### (b) Activity assays for a carbamate hydrolase / urethanase.

- **Colorimetric/fluorometric surrogate substrates.** p-Nitrophenyl carbamate / p-nitrophenyl esters give a 405–410 nm absorbance readout; 4-methylumbelliferyl (4MU) esters give a fluorescence readout (used as the primary screen in Lauko 2025).
- **UMG-SP2 / urethanase-specific assays.** The 2025 RSC *Green Chemistry* colorimetric urethanase assay (DOI 10.1039/D5GC03560K) uses model carbamates synthesized from isocyanate + ethylene-glycol units (1a–1d) and from MDA + chloroformates (1e, 1f); carbamate 1e was used in the original UMG-SP screens. UMG-SP2 model substrates include

BBC [4-hydroxybutyl (4-(4-aminobenzyl)phenyl) carbamate] and dicarbamates such as MDI-DEG (releasing MDA). The JACS 2025 study (DOI 10.1021/jacs.5c13147) uses an asymmetric-carbamate model compound that releases two UPLC-MS-detectable products, probing both esterase and amidase activity. (RSC Publishing) (Wiley Online Library)

- **Product detection.** HPLC/LC-MS for diamine (e.g., MDA) and alcohol/diol release. The *Advanced Science* 2025 engineering paper (DOI 10.1002/advs.202416019) quantified MDA release: variant Q399A released 7.4× more MDA from MDI-DEG, and A141G gave 30.7× improved degradation on polyester-PUR particles by turbidimetry.
- **Kinetics.** Michaelis-Menten fits from initial rates across a substrate-concentration series → kcat, KM, kcat/KM, in 96-well plate-reader format.

### (c) High-throughput screening of many designs.

- **Cell-free / TXTL.** NEB PURExpress (reconstituted PURE system) and Arbor/Daicel myTXTL (*E. coli* lysate) produce protein in ~2 days vs. ~3 weeks in vivo — ideal for rapid triage of designs, including from linear DNA or minicircles. An academic low-cost CFPS formulation has been demonstrated at ~£0.044 per 12 µL reaction; rolling-circle-amplified synthetic DNA has been used to screen carbon-nitrogen-hydrolase superfamily panels cell-free in roughly half the time of cell-based workflows.
- **Plate-based activity screening** on lysate or CFPS reactions using the substrates above.
- **Published hit rates (Lauko et al., *Science* 2025, de-novo serine hydrolases):** fluorophosphonate (FP) probe labeling (folded protein with an activated serine) rose from 3% → 17% (rounds 1→2) and reached 84% in round 3; catalytic-activity hit rates were 1.6% (round 1) and 5.2% (round 2), rising to 73% in a four-state-filtered set; one campaign found 2 active designs out of 20 tested (~10%). 812 designs were the explicitly analyzed set across rounds 1-3 plus prior NTF2 designs. (bakerlab + 3)

### (d) Timeline & cost to validate 10-100 designs.

- *In-house/academic, 10-30 designs:* ~3-6 months, ~\$15-50K (genes + media + consumables + core fees).
- *CRO route:* ProteoGenix *E. coli* expression from ~\$2,160/protein (vendor-listed, not re-verified); GenScript / Sino Biological / Bio Basic per-protein expression-and-purification packages (typically low-thousands each). 100 designs by CRO is \$100K+, so cell-free in-house triage first, then CRO for confirmed leads.

**(e) CROs / enzyme-engineering providers.** GenScript (gene synthesis + bacterial expression + enzyme production), ProteoGenix, Sino Biological, Bio Basic, ProMab. For PU/plastic-degrading expertise the true centers of competence are academic: Bornscheuer & Wei (Univ. Greifswald), Weidong Liu (Tianjin Institute of Industrial Biotechnology, CAS), and Westh & Morth (DTU). Protein Evolution and Samsara Eco are commercial analogs (not service providers).

## PART 2 — Engine A: Polymer / Fiber Wet-Lab Validation Pathway

**(a) Lab-scale synthesis of segmented PU-urea.** Two-step solution polymerization: (1) react excess aliphatic diisocyanate (HDI, H12MDI, or IPDI) with a ~2000 g/mol macrodiol (PCL-diol or polycarbonate-diol) in DMAc/DMF/toluene at 60–85 °C under N<sub>2</sub> with stannous octoate or DBTDL to form an isocyanate-terminated prepolymer at NCO:OH ≈ 1.03–2:1; (2) chain-extend with a short diol/diamine (ethylene glycol, BDO, ethylenediamine, piperazine) to build the segmented (AB)<sub>n</sub> hard/soft architecture. Published exemplars: PCL/HDI/PEG or /EDA two-step solution routes (Lee 2018, Hindawi BMRI); PCL/MDI/BDO solution polymerization; segmented PU-urea shape-memory polymers from PCL-diol/HDI/piperazine. Monomers from Sigma-Aldrich/Millipore, Covestro, Lubrizol; PCL-diol grades from specialty suppliers (e.g., ERA Polymer). **(b) Characterization.** GPC/SEC (Mn/Mw, dispersity); DSC (Tg/Tm, crystallinity); FTIR + <sup>1</sup>H NMR (urethane/urea confirmation, hard-segment content); tensile (modulus, elongation at break); DMA (storage modulus, tan δ). University materials cores charge per-sample/per-hour instrument fees. **(c) Fiber formation.** Dry/solution spinning uses 30–40 wt% dope; wet spinning uses ~20–25 wt% dope into a coagulation bath (demonstrated for PU elastic biofibers at 70–90 μm in DMF/warm-water baths). Electrospinning gives PCL-PU nanofiber mats. **For a research demonstration, wet spinning or solution-cast films/fibers is the most realistic, lowest-capex route;** electrospun mats are excellent for degradation assays because of high surface area. **(d) Enzymatic degradation testing.** Incubate fiber/film with the designed enzyme in buffer at near-neutral pH and controlled temperature with gentle mixing (cf. *Humicola insolens* cutinase protocol: 40–50 °C, pH 8, 6–168 h). Measure: gravimetric **weight loss** over time (normalize to surface area); **monomer/oligomer release** by HPLC-RID/LC-MS (diols, diamine, diacids); GPC for Mn/Mw decrease (HiC reduced Mn by 84%, Mw by 42% on a PU-polyester copolymer); SEM for surface erosion/cracking; FTIR/DSC/TGA on the solid residue. Cascade precedent: HiC + UMG-SP2 produced ~3× more MDA than SP2 alone. **The enzyme-free control is critical** — published work shows negligible abiotic hydrolysis of polyester/PU films in buffer without enzyme. **(e) Timeline & cost.** Polymer synthesis + characterization: ~2–4 months, ~\$20–60K with academic instrument access. Facilities: university polymer/textile labs — NC State Wilson College of Textiles (fiber spinning), Clemson, Georgia Tech — plus specialty contract polymer-synthesis houses and shared-instrumentation cores. (PubMed Central + 4)

## PART 3 — Integrated Co-Design Proof & Academic-Partnership Model

**(a) Proving the matched-pair hypothesis.** Run a 2×2 specificity matrix: {designed enzyme, no-enzyme control} × {designed cleavable fiber, conventional spandex}. The designed enzyme should release stoichiometric monomers from the designed fiber but far less from conventional PU, and the designed fiber should resist a non-matched control enzyme. Quantify selectivity by monomer-release ratio (LC-MS) plus weight-loss and Mn drop. **(b) Minimum viable proof-of-concept (the one experiment).** Express ONE lead designed urethanase → confirm activity on a colorimetric carbamate surrogate → demonstrate measurable monomer release (HPLC/LC-MS) from a small coupon of the designed polymer against a clean enzyme-free negative control. This single "design → expressed enzyme → cleaves designed material → releases monomers" chain de-

risks all three engines simultaneously. **(c) Academic-lab partnerships.** *De-novo enzyme design:* Baker Lab / Institute for Protein Design (UW) and its spinouts/licensees (e.g., Vilya, which licensed the ChemNet/PLACER provisional). *Urethanase / PU-degrading enzymes:* Bornscheuer & Wei (Greifswald), Weidong Liu (Tianjin, CAS), Westh & Morth (DTU). *Recyclable/cleavable PU:* NREL BOTTLE polymer groups. **Mechanism:** sponsored research agreements (SRA) via the university tech-transfer/innovation office; STTR (formal IP agreement required at award); materials-science user facilities. Many universities reduce/waive F&A overhead for Phase I SBIR/STTR (UNC waives F&A; UC Irvine offers a reduced 26% MTDC rate). **(d) US national-lab / user facilities.** The NREL-led DOE **BOTTLE Consortium** ("Bio-Optimized Technologies to keep Thermoplastics out of Landfills and the Environment"; CEO Gregg Beckham; partners include LANL, ORNL, Colorado State, MIT) explicitly pursues "redesign of tomorrow's plastics to be recyclable by design" — directly REWEAR-FUSED's thesis. DOE national labs partner via **CRADA** (with an option to negotiate up to an exclusive license on subject inventions) and Strategic Partnership Projects; non-proprietary research intended for publication is typically free, while proprietary work is on full-cost-recovery. University shared-instrumentation cores cover GPC/DSC/SEM/LC-MS. (OSP + 5)

## PART 4 — IP & Provisional-Patent Pathway

**(a) What's patentable.** A de-novo-designed protein/enzyme is non-naturally-occurring and human-made, generally patentable as a **composition of matter** under 35 U.S.C. §101 (the *Diamond v. Chakrabarty* "anything under the sun made by man" standard; supported by USPTO MPEP 2106). Claims should recite the physical molecule and/or its function — a bare amino-acid sequence as pure information/coordinates risks rejection as "nonfunctional descriptive material"/abstract idea. Also patentable: the **designed polymer composition**; the **co-design method**; and the **matched enzyme+polymer system/pair** (often the strongest, most defensible claim). Precedent: UW/Baker hold granted composition-and-method patents on designed proteins (e.g., US 10,385,333; US 10,920,217); the Lauko paper discloses a provisional (application 63/535,404) covering the ChemNet/PLACER design software, filed by UW and licensed to spinout Vilya — illustrating the method-tool-vs-composition distinction. **(b) Provisional process/cost/timeline.** File via USPTO Patent Center; the filing fee is ~\$60–65 (micro entity), ~\$130 (small entity), ~\$260 (large). Attorney-drafted provisionals run ~\$2,000–6,000 ("all-in" from ~\$2,500). It confers 12 months of "patent pending" and must be converted to a non-provisional within 12 months or it lapses. Micro-entity status ( $\leq$ \$251,190 gross income as of Sept 9, 2025;  $\leq$ 4 prior applications) gives 80% fee reductions; provisional applications don't count toward the application limit and can be filed in unlimited number. **(c) Freedom-to-operate.** Enzymatic PET recycling is heavily patented by **Carbios**, which "owns 116 titles worldwide representing 30 patent families" (six protecting the full recycling method, five on PET-degrading enzymes; ICCG PETase variant; expanding to PU/polyamide). **Samsara Eco** (AI-designed enzymes; a patented dense-plastic pre-processing step; nylon 6/6,6 + polyester; pursuing PU/polyamide). **Protein Evolution** (AI enzymes for PU/nylon, partnered with Basecamp Research). Bio-based/recyclable spandex: **Hyosung** (regen BIO Spandex; sugarcane bio-BDO via Geno; vertically integrated

Vietnam plant, production from April 2026) and **LYCRA** (QIRA bio-BDO via Qore, the Cargill-HELM JV; ~70% bio-derived content). **FTO strategy:** REWEAR-FUSED's white space is *matched co-design* (a purpose-built cleavable fiber + its bespoke enzyme), distinct from incumbents who engineer enzymes to attack *existing* commodity plastics. Commission a professional FTO/landscape search (~\$2,500–3,000) before non-provisional conversion. **(d) Credible "active progress toward IP" for the accelerator.** File the provisional (cheap, fast, real "patent pending") covering the matched-pair method + lead sequences before/around the Cox application; log a dated invention disclosure; if partnering with a university, negotiate an option/license to the relevant background IP. This directly satisfies the Cox Residency eligibility criterion: "Existing patents or active progress toward securing intellectual property licenses." (USPTO + 4)

## **PART 5 — Funding & Timeline to Pilot**

### **(a) Phased timeline & cost.**

- **Phase 0 (now–3 mo):** file provisional; recruit academic PI; register SAM.gov/UEI; submit NSF Project Pitch. ~\$5–15K.
- **Phase 1 (mo 3–12):** express + assay lead enzymes (cell-free triage → IMAC/SEC); synthesize + characterize fiber; run matched-pair MVP. ~\$150–350K (STTR Phase I scale).
- **Phase 2 (mo 12–36):** enzyme engineering for activity/stability; fiber scale-up + spinning; kg-scale enzyme; closed-loop bench pilot. ~\$1–2M (SBIR/STTR Phase II).

**(b) Non-dilutive sources. NSF SBIR/STTR** — under solicitation NSF 26-510 (announced May 26, 2026), the program restarted with \$250M across Phase I/II/Fast-Track plus a \$40M scientific-instrumentation pilot (NSF 26-511) and a Strategic Breakthrough tier up to \$30M; **Phase I up to \$305,000 for 6–18 months** (incl. up to \$6,500 TABA), Phase II up to \$1.25M; mandatory Project Pitch reopened June 2, 2026; full-proposal deadlines July 27, 2026, Nov 4, 2026, and March 4, 2027. STTR requires a research-institution subaward on a 60/40 (small-business/research-institution) work split (≥30% minimum at the institution). **DOE Office of Science SBIR/STTR** (Phase I historically up to ~\$225K) plus DOE BOTTLE/bioenergy programs. USDA SBIR (bio-based), EPA grants. **Textile-circularity:** Fashion for Good innovation programmes (Laudes Foundation-backed); the Global Fashion Agenda × PDS Ventures **Trailblazer Programme** (PDS Ventures invests USD 50,000–200,000; EVERBLOOM, a protein-engineering fiber startup, won the inaugural 2024 edition; Refiberd 2025; Synflux 2026); IndieBio/SOSV (~\$250K + lab space); MIT Solve; EIC Accelerator (EU, up to €2.5M grant + equity). (NSF - National Science Fou...)

**(c) What the Cox Cleantech Residency enables.** \$10K stipend, 6 weeks in-person in Atlanta, equity-free; weekly 1:1s, mentor swarms, investor-swarm pitch events, warm VC/angel intros, and pilot access via Cox corporate partners (the Carter's children's-apparel circularity track). Realistically it funds the provisional filing plus a small batch of synthetic genes/assays, and—most valuably—provides the network and a pilot-customer pathway, plus a credible bridge into the larger **Cox Cleantech Accelerator** (12 weeks, \$250K investment). Use the residency to (i) lock the

provisional, (ii) sign an academic LOI/SRA term sheet, (iii) submit the NSF Project Pitch, and (iv) line up a brand pilot LOI. (Residency applications close June 28, 2026; demo day June 17.)

(d) 12- and 36-month roadmaps.

- **12-month:** provisional filed → academic STTR partner secured → NSF Phase I awarded → matched-pair MVP data + ≥1 brand pilot LOI.
- **36-month:** engineered enzyme with industrially relevant kcat/KM on the designed substrate; scaled fiber + spinning demonstration; kg-scale enzyme production; validated closed-loop bench pilot; non-provisional + PCT filed; SBIR/STTR Phase II + seed round.

Budget & Timeline Table (today → pilot)

| Phase               | Months | Key activities                                                                                                                                           | Cost (USD) | Funding source                                 |
|---------------------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------|------------|------------------------------------------------|
| 0 — Setup           | 0-3    | Provisional patent; recruit academic PI; SAM.gov; NSF Project Pitch                                                                                      | \$5-15K    | Cox Residency \$10K + founder cash             |
| 1 — MVP proof       | 3-12   | Cell-free triage of designs; express + purify lead urethanase; carbamate assay; synthesize + characterize fiber; matched-pair monomer-release experiment | \$150-350K | NSF/DOE STTR Phase I (≤\$305K) + textile prize |
| 2 — De-risk & scale | 12-36  | Enzyme engineering (activity/stability); fiber scale-up + wet spinning; kg enzyme; closed-loop bench pilot; non-provisional + PCT                        | \$1-2M     | SBIR/STTR Phase II (≤\$1.25M) + seed           |

Recommendations

1. **Run the MVP matched-pair experiment first** (express 1 lead urethanase → carbamate colorimetric assay → monomer release from designed coupon vs. enzyme-free control). *Advance threshold:* statistically significant, dose-dependent monomer release above control by LC-MS. If it fails, iterate enzyme designs cheaply via cell-free TXTL before any fiber scale-up spend.
2. **Secure an academic STTR partner now** — an IPD/Baker-adjacent group for enzymes, a polymer/textiles lab for fiber, and NREL BOTTLE for recyclable-by-design framing — and **file the NSF Project Pitch for the July 27, 2026 cycle.**

3. **File a micro-entity provisional** (~\$60-130 USPTO + ~\$2-4K attorney) **on the matched-pair method + lead sequences before the Cox application closes (June 28, 2026)** to satisfy the IP eligibility criterion.
4. **Triage designs by cell-free expression** before committing to full *E. coli* purification — this mirrors the Baker-lab low-throughput strategy that found hits in as few as 20 designs and keeps first-pass cost in the hundreds-of-dollars range.
5. **Stage the funding ladder:** Cox Residency → NSF/DOE STTR Phase I → textile-circularity prize/accelerator (Trailblazer, Fashion for Good, IndieBio) → SBIR/STTR Phase II + seed.

## Caveats

- **Activity-class gap:** the de-novo hit rates and the  $2.2 \times 10^5 \text{ M}^{-1}\text{s}^{-1}$  figure (Lauko 2025) are for serine esterase hydrolysis, not carbamate/urethane hydrolysis. Urethane cleavage is harder, and current best urethanases (UMG-SP2 and engineered variants) have far lower activity on real PU. Present this to judges as the central engineering target, not a solved problem. Note also a version discrepancy: the bioRxiv preprint reported  $3.8 \times 10^3 \text{ M}^{-1}\text{s}^{-1}$ ; the peer-reviewed *Science* paper reports  $2.2 \times 10^5$  after later design rounds — cite the published figure. (nih)
- **Marketing vs. data:** several competitor claims (Samsara, Protein Evolution PU work) come from company/trade press, not peer-reviewed data; treat "infinite recycling" and scale-up language accordingly.
- **Moving targets:** USPTO fees, NSF SBIR amounts/dates, and vendor prices are current as of mid-2026 but change — verify before filing/submitting. Twist per-bp pricing, the UNC core fee, and the ProteoGenix \$2,160 figure are vendor-listed estimates not independently re-verified here.
- **Program distinction:** the Cox "\$10K, 6-week" residency and the "\$250K, 12-week" accelerator are separate programs; the residency is the equity-free bridge referenced in the task.
- **Mass-fraction framing:** elastane is only ~2% of a typical stretch garment; the circularity value proposition should emphasize that a cleavable elastane enables recycling/separation of the *whole* blend, not just the spandex itself — the most compelling pitch for the Carter's children's-apparel track.